

Natural vs synthetic

A frog sits inside a water droplet on a lotus leaf in a pond in Nepal. Lotus leaves are known for their hydrophobic (water-repellent) properties, which stop water from sinking into the leaves. Instead, it forms spherical droplets on the surface. The attractions between the water molecules in these droplets are so strong, they can trap tiny creatures like this frog. The bumpy surface of the human-made material on the right has been designed to mimic that of the lotus leaf to make it waterproof.

The lotus leaf
has also inspired
the design of
self-cleaning
paint!



BUMPY SURFACE

Water does not wet a lotus leaf because the leaf surface is covered in tiny, pointy structures and coated in natural waxes. The ridged surface reduces how much of the leaf is in contact with the water.